not be numbers. For example in the random experiment of tossing a coin, the outcomes are H (head) or T (tail). It is necessary to deal with numerical values, in some situation, for outcomes of random experiment. Therefore, we assign a number to each outcome of the experiment say 1to head and 0 to tail. Such an assignment of numerical values to the elements in S is called a *random variable*. *A random variable is a function*. Thus, a random variable is:

Definition 11.1

A random variable X *is a function defined on a sample space* S *into the real numbers* $\mathbb{R}$ *such that the inverse image of points or subset or interval of* $\mathbb{R}$ *is an event in* S, *for which probability is assigned.*

We use the capital letters of the alphabet, such as *X, Y,* and Z to represent the random variables and the small letters, such as *x, y,* and z to represent the possible values of the random variables.

Suppose $S = \{\omega_1, \omega_2, \omega_3, \ldots\}$ is the sample space of a random experiment and $\mathbb{R}$ denotes the real line. Then the random variable X is a real valued function defined on S and is denoted by $X : S \to \mathbb{R}$. If ω is a sample point in S, then $X(\omega)$ is a real number.

The range set is the collection of $X(\omega)$ such that $\omega \in S$. That is the range set denoted by R_x is $R_x = \{X(\omega) / \omega \in S\}$.

Fig 11.1 shows the mapping of some sample points ω_i or events of the *Sample space* S on the *real line R*.

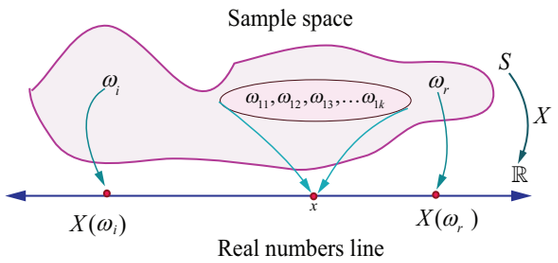


Fig. 11.1

For instance, if x is a possible value of X for $\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k} \in S$, then $\{\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k}\}$ is called inverse image of x.

That is $X^{-1}(x) = \{\omega_{11}, \omega_{12}, \omega_{13}, \ldots \omega_{1k}\}$ is an event in S

Illustration 11.1

Suppose a coin is tossed once. The sample space consists of two sample points H (head) and T (tail).

That is $S = \{T, H\}$

Let $X : S \to \mathbb{R}$ be the number of heads

Then $X(T) = 0$, and $X(H) = 1$.

Thus X is a random variable that takes on the values 0 and 1. If $X(\omega)$ denotes the number of heads, then

$$X(\omega) = \begin{cases} 0 & \text{for } \omega = \text{Tail} \\ 1 & \text{for } \omega = \text{Head} \end{cases}$$

Example 11.1

Suppose two coins are tossed once. If X denotes the number of tails, (i) write down the sample space (ii) find the inverse image of 1 (iii) the values of the random variable and number of elements in its inverse images.

Solution

(i) The sample space $S = \{H, T\} \times \{H, T\}$

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